TITLE : SULAM PROJECT PROPOSAL

SUBJECT: WIX2001

GROUP: OCC1, TEAM 4

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TABLE OF CONTENTS

1.0 Overview

The lack of recognition and manpower, specifically low public engagement and limited volunteer support is severely threatening the survival and quality of our local heritage games. Our proposed VisionARies seeks to create a solution to this problem by using modern technology to enhance these games without removing its historical elements. We are seeking to create an Augmented-Reality guide for the tourists. By implementing AR and using TikTok as our platform, we are able to create a more engaging approach to the tourist to play our local heritage games at the tip of their fingers. We are aiming to dramatically increase public awareness and generate a more passionate participation while securing the future of our cultural treasures.

2.0 Collaborator

2.1 Organization

Badan Warisan Malaysia (BWM) will serve as the primary cultural and heritage collaborator for this project. Through this partnership, Malaysia's traditional heritage games can be accurately represented by our AR application while maintaining authenticity, cultural sensitivity and historical integrity.

2.2 Roles and Responsibility of Collaborator

1) **Cultural Expertise & Content Validation**
   a) Provide proficient knowledge on traditional Malaysian heritage games.
   b) Validate all AR content game mechanics and narrative elements to ensure cultural accuracy.
   c) Review prototypes to ensure that the AR representation aligns with heritage values and practices.
2) **Advisory Role in Heritage Compliance**
   a) Advise on cultural protocols to ensure AR project respect heritage values and do not misrepresent traditional practices.
   b) Ensure that the project aligns with national heritage preservation guidelines.

2.3 Benefits to the Collaborator

This AR project benefits BWM by providing digital preservation tools that strengthen their policy advocacy efforts. It serves as an evidence that technology can effectively protect intangible heritage, supports youth engagement, raise public awareness and offers a scalable model for future heritage initiatives. These outcomes help BWM push for policies that foster a conservation-friendly national environment.

2.4 Mode of Collaboration

The collaboration will follow a consultative and co-development structure, which includes:

- Monthly coordination emails

- Content review cycles at every development milestone.

2.5 Expected Commitment

1) BWM will provide advisory and content validation support throughout the development period.
2) The project development team will take full responsibility for technical implementation and AR design.

3.0 Target Group

The target audience for this project is tourists at BWM regardless of their age, gender, or race, who are interested in exploring traditional games. Mobile phones are required for them to view the traditional games simulation since it is accessible through QR codes. The AR technology is designed to help them gain a complete understanding of how to play the traditional games in a fun, easy, and accessible way.

4.0 Problem and Needs Identification

4.1 Problem Analysis

The current operational framework utilized by Badan Warisan Malaysia (BWM) to manage its heritage sites are limited to significant constraints, primarily due to the lack of technological infrastructure, human resources, and overall visitor experience. These constraints act as a bottleneck for BWM to scale their cultural preservation and to meet their educational goals.

Core Problem Statement

BWM's heritage sites face a challenge concerning visitor engagement and operational resource management. This is particularly caused by guiding methods that are conventional and manpower-intensive, and the limitations of a static physical exhibit.

The existing framework introduces several key challenges:

a) Operational Inefficiency
The lack of guiding manpower makes it difficult for staff to cater to multiple visitors at once because of a low guide-to-visitor ratio.

b) Risk of Inconsistent Knowledge Delivery

Guides are required to repeat the same knowledge-sharing dialogues multiple times daily. This increases the risk of cognitive fatigue which could lead to an inconsistent flow of information and the exclusion of critical information.

c) **Limitation of Spatial and Temporal Contexualization**
The reality of the historical and cultural site is limited to what is physically available present-day and what the guides present to the visitors. This limits the visitor's potential to comprehend the historical environment that is now unavailable.

4.2 Identified Needs and Strategic Objectives

To tackle the aforementioned problems, BWM needs a solution that:

- **Enhances operational efficiency and assists the guide in knowledge delivery.**
- **Reconstructs historical content digitally to replace what is not available present-day.**
- **Acts as a bridge between the visitors and the heritage site so that there is an increased percentage of interaction and personal connection.**

5.0 Project Objectives

The objective of this project is to create an Augmented Reality (AR) heritage game for BWM to enhance visitor engagement while improving operational efficiency. This project aims are:

a) **Reduce the Workload of Heritage Site Staff**

By applying AR in building a tutorial or gamified platform to introduce heritage games to visitors, the system minimizes the need for constant and repetitive explanation of staff especially during peak hours. Other than that, automated AR experience ensures consistent explanation rather than inconsistent guides that may arise from guide fatigue or repeated manual experience.

b) **Increase Visitor Awareness and Understanding of Cultural Heritage**

The project aims to increase visitor awareness using AR tutorials to present heritage games in a simple, easy to understand and in visually enriched manner. In addition, the AR experience strengthens appreciation for local heritage by helping visitors to better understand the unique culture, rules and history behind each game.

c) **Enhance Visitor Engagement Through Interactive Learning**

To create an interesting and interactive learning experience, visitors are encouraged to actively participate and apply their understanding of the heritage games by playing the game in real life. Meanwhile, the hands-on involvement fosters deeper interest and builds a personal connection between visitor and the heritage site by engaging in enjoyable activities, especially playing heritage games with friends and family. Visitors can create meaningful and memorable experiences that strengthen their appreciation of local culture.

6.0 Proposed Activities

6.1 Augmented Reality (AR) Heritage Game Development

This project focuses on developing an AR application designed to educate the audience on Malaysian heritage games. This serves as an interactive demonstration platform to teach audiences the rules and mechanics of these games.
Combination of the following methods for engagement and clarity:

a) Visual Step-by-Step Instructions
b) Demonstration Videos
c) 3D Animated Models

6.2 Social Media Filter for TikTok

Proposes the creation of TikTok filters to promote interaction with Malaysian heritage games among modern audiences.

The proposed filter will aim for an interactive experience that allows users to virtually engage with the game elements. The fallback will be to utilize the filter as an accessible educational tool by:

a) Displaying instructional videos of the gameplay.
b) Showcasing photos demonstrating the game setup and key steps.

We kindly request your team review both options and select the one that best aligns with the targeted audience.

7.0 Group Member and Job Role

Role	Name
Project Manager/ Team Lead	Nur Sahira Batrisyia binti Muhammad Daud
Documentation Lead	Nurin Allya Insyira binti Norshamsulazman
AR Developer	Muhammad Hariz bin Shamsudin Kearney Leith Julius Edlan Nabil bin Akmal Hamdi
UI/UX/3D Model Designer	Najah Safwah binti Monarizzal Anis Atirah binti Mohd Aris
Multimedia Experience Architect	Muhammad Aqlan bin Azizul
Research & Content Lead	Garry Fresco Simon
Testing Coordinator	Ameer Zafran bin Mohd Abdullah Shabri

Role Descriptions

Role	Role Description
Project Manager/ Team Lead	The Project Manager/Team Lead defines the project scope, sets milestones and timelines, manages the budget and resources, and acts as the primary point of contact for stakeholders. They lead the team, manage communication, mitigate risks, and ensure the project remains on track and within specified constraints, facilitating a smooth workflow between all other roles.
Documentation Lead	The Documentation Lead is responsible for creating, organizing, and maintaining all project-related records. This includes technical documentation , design specifications, user guides, and project records. Their primary goal is to ensure information is easily accessible, clear, and up-to-date, facilitating onboarding, knowledge transfer, and maintenance.

AR Developer	AR Developers are responsible for writing the code that brings the AR experience to life. This includes implementing AR tracking, integrating 3D models and UI elements, handling sensor data, and ensuring the application performs well across target devices.
UI/UX/3D Model Designer	UI/UX/3D Model Designer are responsible for creating: a) UI (User Interface): The visual elements users interact with. b) UX (User Experience): Ensuring the app is intuitive, efficient, and enjoyable to use, both in 2D menus and within the 3D AR space. c) 3D Models: Creating, optimizing, and texturing the three-dimensional assets that populate the augmented reality experience.
Multimedia Experience Architect	The Multimedia Experience Architect focuses on the overall immersive and sensory quality of the AR application. They define how various non-coding/non-modeling elements such as sound design, music, haptic feedback, animation principles, and transitional effects are integrated to create a cohesive and engaging user journey. This role bridges the gap between the design and development teams to ensure a high-fidelity, polished, and memorable final product.
Research & Content Lead	The Research & Content Lead defines the factual accuracy, narrative structure, and overall message of the AR experience. They research required topics, draft text, scripts, or content descriptions, and ensure all displayed information is clear, engaging, and appropriate for the target audience.
Testing Coordinator	The Testing Coordinator ensures the application is stable, functional, and meets all requirements before release. They are responsible for developing the test plan, defining test cases (functional, performance, usability, AR tracking accuracy), coordinating internal and external testing efforts, tracking reported bugs, and working with the developers to verify fixes.

8.0 Expected Outcome and Impact on Society And Group

8.1 Expected Outcomes (KPI and Deliverables)

The VisionARies project is designed to deliver tangible, measurable outcomes that directly resolve the identified constraints of low recognition and operational inefficiency at BWM heritage sites.

Category	Expected Outcome	KPI	Target (6 months after deployment)
i. Public Recognition & Engagement	Significant increase in awareness and digital interest in heritage games.	- TikTok Engagement Rate - AR activation (Unique scans of QR marker)	50% increase in average engagement rate across TikTok. 30% increase in site visitor engagement (QR scans)
ii. Operational Efficiency	Reduction in the manual workload required for repetitive game instruction.	Staff Time Allocation (Reduction in time spent by BWM staff explaining basic game rules and history)	25% reduction in guide time dedicated to introductory explanations during peak hours.
iii. Preservation & Manpower	Secure digital archive and a foundation for new volunteer recruitment.	Digital Asset Completion: Percentage of identified heritage games successfully digitized into the AR platform. Volunteer inquiry rate	100% of selected games are documented and digitized.

8.2 Impact on Society and Group

The successful achievement of these outcomes will generate a significant, multifaceted impact on both BWM and the wider Malaysian society by leveraging technology for cultural direction.

The impact on Badan Warisan Malaysia (BWM) and Cultural Groups
VisionARies provides BWM with a sustainable digital tool for fulfilling its mandate of heritage preservation. The AR guide serves as a permanent, consistent digital steward, ensuring that the knowledge delivered is accurate, comprehensive, and resilient to staff fatigue or turnover. This project solidifies BWM's reputation as an innovative institution that embraces modern technology to protect intangible cultural heritage. By reducing the

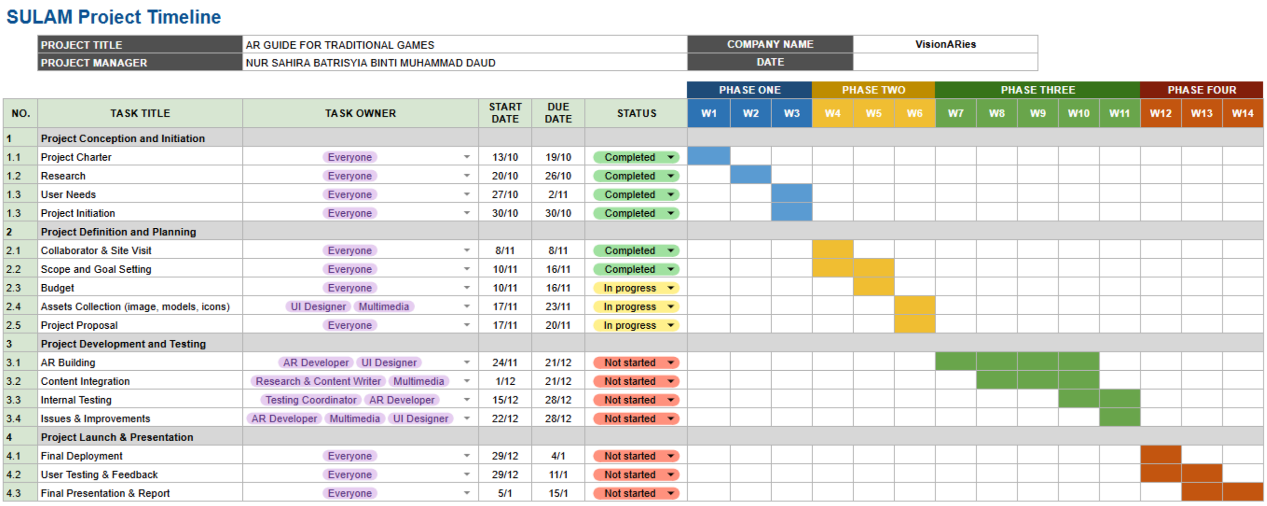
manual burden, BWM staff can dedicate more time to high-value activities, such as advanced research, specialized tours, and community outreach. This project secures the survival and quality of the local heritage games by transforming them from static exhibits into engaging, interactive cultural practices.

Impact on Society and Youth Engagement

This project profoundly influences the relationship between **Malaysian Youth** and their cultural history. By using TikTok, VisionARies effectively lowers the barrier to entry for heritage learning. It translates complex historical context into a visually rich and interactive experience, sparking curiosity and moving young individuals from being passive observers to active cultural participants. The resulting public awareness fosters a stronger sense of national identity and cultural pride, ensuring that these heritage games remain a vibrant, living part of Malaysian society.

9.0 Project Timeline

The development of our project is planned across fourteen (14) weeks and divided into four major phases: **Initiation**, **Planning**, **Development**, and **Launch**. Each phase contains specific tasks necessary to ensure the successful completion of the prototype. The following sections describe the activities involved in each phase.



SULAM Project Timeline

PROJECT TITLE	AR GUIDE FOR TRADITIONAL GAMES	COMPANY NAME	VisionARies
PROJECT MANAGER	NUR SAHIRA BATRISYIA BINTI MUHAMMAD DAUD	DATE	

						PHASE ONE			PHASE TWO			PHASE THREE					PHASE FOUR		
NO.	TASK TITLE	TASK OWNER	START DATE	DUE DATE	STATUS	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
1	**Project Conception and Initiation**																		
1.1	Project Charter	Everyone	13/10	19/10	Completed														
1.2	Research	Everyone	20/10	26/10	Completed														
1.3	User Needs	Everyone	27/10	2/11	Completed														
1.3	Project Initiation	Everyone	30/10	30/10	Completed														
2	**Project Definition and Planning**																		
2.1	Collaborator & Site Visit	Everyone	8/11	8/11	Completed														
2.2	Scope and Goal Setting	Everyone	10/11	16/11	Completed														
2.3	Budget	Everyone	10/11	16/11	In progress														
2.4	Assets Collection (image, models, icons)	UI Designer, Multimedia	17/11	23/11	In progress														
2.5	Project Proposal	Everyone	17/11	20/11	In progress														
3	**Project Development and Testing**																		
3.1	AR Building	AR Developer, UI Designer	24/11	21/12	Not started														
3.2	Content Integration	Research & Content Writer, Multimedia	1/12	21/12	Not started														
3.3	Internal Testing	Testing Coordinator, AR Developer	15/12	28/12	Not started														
3.4	Issues & Improvements	AR Developer, Multimedia, UI Designer	22/12	28/12	Not started														
4	**Project Launch & Presentation**																		
4.1	Final Deployment	Everyone	29/12	4/1	Not started														
4.2	User Testing & Feedback	Everyone	29/12	11/1	Not started														
4.3	Final Presentation & Report	Everyone	5/1	15/1	Not started														

Figure 1: Gantt chart for project timeline

9.1 Phase 1: Project Conception and Initiation (Week 1–3)

This phase focuses on understanding what we want to build and why.

Key Activities:

- Prepare the project charter and define scope.
- Conduct initial research on AR and traditional games.
- Identify user needs and expectations.
- Finalize project direction with the team.

9.2 Phase 2: Project Definition and Planning (Week 4--6)

This phase ensures that all necessary preparations are in place before development begins.

Key Activities:

- Meet collaborators and conduct site visit.
- Finalize project goals, features, and scope.
- Plan the project budget.
- Collect required images, icons, and multimedia assets.
- Prepare and complete the project proposal.

9.3 Phase 3: Project Development and Testing (Week 7–11)

This is the execution phase where the AR prototype is constructed and refined. This is the most critical phase in our project.

Key Activities:

- Develop the AR prototype and game demonstrations.
- Integrate content such as rules, visuals, and audio.
- Conduct internal testing to identify issues.
- Fix bugs and refine the overall user experience

9.4 Phase 4: Project Launch and Presentation (Week 12–14)

This phase focuses on final deployment and presentation of the project outcome.

Key Activities:

- Finalize and deploy the AR prototype.
- Conduct user testing and gather feedback.
- Present the final product and submit the project report.

The timeline ensures that all tasks are organized and completed on schedule, enabling the team to deliver a functional AR prototype within the 14-week project period.

10.0 Conclusion

In conclusion, the VisionARies project aims to bridge the gap between tradition and modern innovation by revitalizing Malaysian heritage games through Augmented Reality. By collaborating with Badan Warisan Malaysia (BWM), this initiative not only supports cultural preservation but also introduces an engaging and accessible learning platform for tourists and the general public. Through AR-guided experiences, visitors gain a consistent, interactive, and meaningful understanding of our heritage while reducing the operational burden on site staff.

This project reflects our commitment to safeguarding intangible cultural heritage by leveraging technology to spark curiosity, strengthen appreciation, and encourage participation among younger generations and international visitors. Ultimately, VisionARies serves as a sustainable digital solution that promotes cultural continuity, enhances visitor engagement, and contributes to long-term heritage preservation efforts in Malaysia.